

AI-Driven Glaucoma Detection: A Deep Learning Approach with Explainable Diagnostics

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Abstract—Glaucoma is one of the primary causes of irreversible blindness worldwide, emphasizing the importance of early and accurate diagnosis. Conventional glaucoma screening methods rely extensively on the expertise of ophthalmologists, making the process subjective, time-consuming, and prone to variability. To overcome these limitations, this study proposes an automated glaucoma detection system based on deep learning techniques for the analysis of retinal fundus images. Transfer learning is employed using state-of-the-art pretrained models, including ResNet50, VGG16, and EfficientNet. The performance of these models is optimized through comprehensive hyperparameter tuning, incorporating variations in learning rates, batch sizes, and the use of modern optimization algorithms such as AdamW. To enhance model explainability, Grad-CAM and Class Activation Mapping (CAM) are integrated, allowing visualization of critical image regions contributing to the prediction. Additionally, a GPT-based image model is incorporated to generate textual insights, improving the interpretability of results. Experimental evaluations compare the performance of different models and optimization techniques, ensuring a robust and clinically relevant AI-driven diagnostic system. Experimental results show that the proposed model achieved a higher classification accuracy and stability in different pretrained architectures. The findings highlight the potential of deep learning in automating Glaucoma screening, enhancing diagnostic accuracy and also shows strong potential for supporting scalable and transparent glaucoma screening in clinical settings.

Index Terms—Glaucoma detection, deep learning, transfer learning, retinal fundus images, ResNet50, EfficientNet, VGG16, AdamW optimizer, Grad-CAM, Class Activation Mapping (CAM), GPT-based image analysis, explainable AI

I. INTRODUCTION

Undiagnosed or untreated glaucoma, which develops into optic neuropathy, results in permanent vision loss. The disease affects the optic nerve, primarily due to elevated intraocular pressure (IOP) or other vascular and neurodegenerative factors. Conventional diagnostic methods, such as tonometry for measuring IOP, visual field testing, and OCT, require specialized equipment and trained ophthalmologists, making large-scale screening difficult, especially in resource-limited settings. Additionally, these diagnostic techniques can exhibit inter-observer variability, further complicating accurate diagnosis.

Given the growing prevalence of glaucoma, particularly among aging populations, there is a critical need for an automated, scalable, and efficient diagnostic approach that can complement clinical decision-making and facilitate early intervention.

Medical imaging has undergone a revolution because to recent developments in deep learning and artificial intelligence (AI), which have made it possible to diagnose diseases automatically and accurately [12]. CNNs, have shown impressive results in the analysis of intricate visual data, including retinal fundus images, which contain crucial structural information about the optic disc, retinal nerve fiber layer, and other indicators of glaucomatous damage. Among various deep learning architectures, pretrained models such as ResNet50, VGG16, and EfficientNet have been widely adopted due to their strong feature extraction capabilities. By leveraging transfer learning, these models can be fine-tuned on medical image datasets, significantly improving performance with limited labeled data. However, while deep learning models can achieve high accuracy, a major challenge in their clinical adoption is their "black box" nature, where decisions are made without clear interpretability. This lack of transparency is a critical barrier in healthcare applications, where explainability and trustworthiness are paramount. To address these challenges, integrating explainability techniques such as Grad-CAM and CAM has become essential. By emphasizing the areas of a picture that have the most influence on the model's choice, these techniques offer pictorial solutions., offering valuable insights for ophthalmologists. In addition to visual explanations, incorporating natural language processing (NLP) models, such as a GPT-based system, can further improve interpretability by generating textual insights that describe retinal abnormalities, providing a more comprehensive analysis for clinicians. The fusion of deep learning with NLP-based image analysis represents a novel approach that enhances both accuracy and accessibility in AI-driven diagnostics. Several automated glaucoma detection systems which are based on deep learning was proposed in recent literature that demonstrating reliable performance in classifying images for clinical decision support [10].

Another crucial aspect of this study is optimizing the model's performance through systematic hyperparameter tuning. Hyperparameters have a significant impact on deep learning models such as learning rates, batch sizes, and optimization algorithms. Selecting the right combination of these parameters is vital for achieving optimal accuracy and training stability. The speed and resilience of learning are impacted by optimizers like Adam, AdamW, and Stochastic Gradient Descent (SGD), which are essential for model convergence. Evaluating different configurations allows for a comparative analysis of their impact on glaucoma detection, ensuring that the best-performing model is selected for clinical implementation. Given that healthcare applications demand both high precision and recall, fine-tuning these parameters ensures a balanced approach, minimizing false negatives that could delay treatment and false positives that could lead to unnecessary anxiety and additional testing.

Annotated datasets is another key factor influencing model performance. Datasets such as REFUGE and ORIGA, which contain expertly labeled retinal fundus images, facilitating supervised learning for glaucoma detection. Preprocessing techniques are applied to standardize the input data and improve model generalization. Data augmentation is applied to improve data diversity and minimize overfitting. By training on diverse datasets, The goal of the suggested method is to guarantee dependability in actual clinical settings by achieving strong performance under a range of retinal imaging circumstances. Experimental evaluations in this study focus on several key performance metrics. These metrics provide a model for distinguishing glaucomatous from non-glaucomatous eyes. The integration of the AdamW optimizer further enhances training stability, preventing overfitting and ensuring smoother convergence. The use of Grad-CAM visualizations confirms that the model accurately focuses on clinically relevant regions and reinforcing.

In addition to technical performance, the practical deployment of AI-based glaucoma detection systems presents several challenges, including regulatory approval, integration with existing ophthalmology workflows, and patient acceptance. Trust in AI-assisted diagnostics remains a crucial concern, as clinicians require confidence that these models can provide reliable and interpretable results. The incorporation of explainability tools addresses some of these concerns, making AI-based glaucoma screening more transparent and clinically acceptable. Furthermore, as AI-driven diagnostic tools become more prevalent, it is essential to ensure that these models are unbiased and generalizable across different populations. Addressing potential biases in training data and validating performance across diverse ethnic and demographic groups will be critical for ensuring equitable healthcare outcomes.

The broader implications of this research extend beyond glaucoma detection. The methodologies explored in this study—transfer learning, model explainability, and NLP-based image analysis—can be applied to various other ophthalmic and medical conditions, contributing to the advancement of AI in healthcare.

As deep learning continues to evolve, integrating multimodal imaging, such as combining fundus photography with OCT

scans, to further improve diagnostic accuracy. Additionally, real-time deployment of these models in telemedicine platforms could expand access to early glaucoma screening, particularly in underserved regions where specialist care is limited. In order to improve patient outcomes and lessen the burden of glaucoma-related blindness worldwide, our research attempts to close the disparity between clinical applicability and technology breakthroughs by continually enhancing AI-based diagnostic tools.

II. RELATED WORK

Recent advancements in automated glaucoma screening and identify gaps that the proposed system addresses here.

A. Deep Learning in Glaucoma Detection

Deep learning has revolutionized medical imaging, particularly in ophthalmology. Convolutional Neural Networks (CNNs) can learn hierarchical picture representations, they have been frequently used for glaucoma diagnosis. Studies have shown that CNN-based models can outperform traditional machine learning methods by extracting high-level features directly from retinal fundus images [1]. However, a major limitation of early CNN-based approaches was their dependency on large labeled datasets, which are often difficult to obtain [11]. Engineers have used transfer learning to get around this problem, using pretrained models like ResNet50, VGG16, and EfficientNet to increase detection accuracy with less labeled examples. [2]. Recent studies demonstrated ensemble approaches that combines CNN, ResNet50 and MLP architectures that improves classification performance in detection tasks [8].

B. Transfer Learning for Medical Picture Analysis

Transfer learning, which enables models built on massive datasets to be tailored for particular medical applications, has been thoroughly investigated in the field of medical picture categorization. Recent research has shown that using pretrained networks, like ResNet and EfficientNet, to detect glaucoma considerably enhances performance when as opposed to creating new models from scratch. [3]. Researchers have achieved advanced outcomes in glaucoma diagnosis by fine-tuning these models and improving hyperparameters including learning rates, batch sizes, and optimizers like AdamW [4]. The effect of various transfer learning models and optimization strategies on detection performance is further examined in this work [15].

C. Explainable AI for Medical Imaging

A critical barrier to clinical adoption of AI systems lies in their opaque decision-making processes, which erode trust among healthcare providers who demand clarity in diagnostic rationale. To address this, post-hoc interpretability methods like Grad-CAM and CAM generate heatmap visualizations identifying anatomical areas in fundus imagery most influential in predicting glaucoma severity. By delineating pathological features such as optic disc changes or retinal nerve fiber layer thinning [14], these tools enable clinicians to audit and validate algorithmic outputs, bridging the gap between AI diagnostics and clinical workflows [5]. Building on prior evidence of Grad-CAM's efficacy in medical imaging [6], our framework combines Grad-CAM and CAM to deliver granular, clinically actionable insights, fostering confidence in automated glaucoma assessment.

D. Integration of GPT-Based Models in Medical Imaging

New opportunities for medical imaging analysis have been made possible by recent developments in large-scale language models (LLMs). GPT-based models have been explored in healthcare applications, offering automatic textual insights and explanations for complex medical images. Research has shown that combining vision models with natural language processing (NLP) techniques can improve diagnostic accuracy and accessibility [7]. The integration of GPT-based models with deep learning for glaucoma detection is a novel approach, allowing automated textual descriptions of retinal fundus images to assist clinicians in decision-making. This study explores the effectiveness of GPT-based analysis in complementing image-based glaucoma detection.

E. Gaps and Research Contributions

Deep learning-based glaucoma detection, several critical limitations persist in existing studies. Many approaches rely heavily on multimodal imaging data, such as OCT and RNFL measurements [1], [4], which restricts their applicability in resource-constrained clinical settings where such devices are unavailable. Additionally, multimodal and hybrid architectures significantly increase computational complexity, demanding high-end hardware and limiting scalability for real-time clinical deployment [1], [4], [6], [11].

Several studies emphasize improved classification accuracy but provide limited clinical interpretability, reducing trust and acceptance among ophthalmologists [3], [11]. Although recent efforts incorporate explainable AI techniques such as LIME or basic visualization methods [5], comparisons with advanced explainability approaches (e.g., Grad-CAM and CAM) remain insufficient. Transfer learning-based approaches have demonstrated performance gains; however, most evaluations are conducted on single or limited datasets, raising concerns regarding generalization across diverse populations, imaging devices, and acquisition conditions [2], [3], [5].

Many studies focus on isolated tasks such as optic disc segmentation rather than delivering an end-to-end diagnostic framework suitable for clinical decision support [2]. Recent transformer-based and capsule-network models achieve promising accuracy but suffer from high training complexity, dependency on large annotated datasets, and inadequate explainability, hindering real-world clinical adoption [11].

Motivated by these gaps, the proposed work introduces an interpretable, computationally efficient glaucoma detection framework that integrates transfer learning, advanced explainable AI techniques, and GPT-based textual analysis. Unlike prior studies, the proposed system emphasizes clinical interpretability, generalization, and practical applicability, while adhering to standardized evaluation protocols highlighted in the JustRAIGS challenge [9].

III. SYSTEM ARCHITECTURE

The proposed system for automated glaucoma detection and analysis follows a modular design, ensuring efficiency,

scalability, and explainability. It consists of multiple sequential stages, starting from data acquisition to final decision-making, providing a comprehensive AI-driven diagnostic tool for ophthalmologists. The overall system architecture is illustrated in Fig. 1 that contain the following phases:

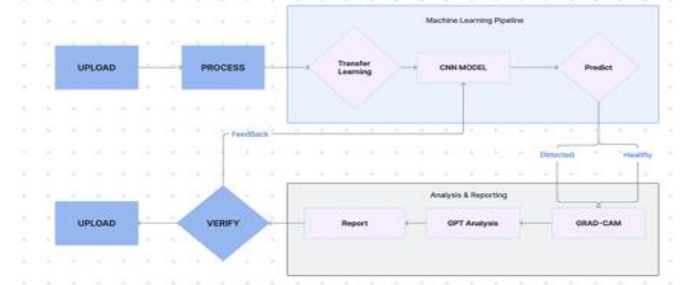


Fig. 1. Architecture of the proposed system

A. Data Acquisition and Preprocessing

The system begins with collecting retinal fundus images from publicly available datasets, such as REFUGE and ORIGA, ensuring a diverse representation of glaucomatous and non-glaucomatous eyes. Data preprocessing is essential to improve model accuracy and involves grayscale conversion, contrast enhancement, noise reduction, and image normalization. These steps ensure that the input images are standardized for deep learning models, reducing variations caused by different imaging conditions and enhancing feature extraction.

B. Model Selection and Transfer Learning

Our methodology employs transfer learning with advanced CNN architectures to harness pretrained feature representations. Architectures including ResNet50, VGG16, and EfficientNet are prioritized due to their proficiency in capturing detailed patterns within retinal fundus images. Each architecture undergoes systematic refinement through hyperparameter tuning, such as optimizing learning rates, batch sizes, and choice of optimizer (Adam, AdamW, SGD). This process enhances model generalizability while maintaining computational efficiency, enabling robust classification performance with limited training iterations.

C. Training and Hyperparameter Optimization

During training, extensive experiments are conducted to determine the best-performing model configurations. The system evaluates different loss functions, including Binary Cross-Entropy, to optimize classification between glaucomatous and non-glaucomatous images. The training process employs data augmentation techniques to increase dataset diversity, preventing overfitting and improving model robustness. Early stopping mechanisms and learning rate schedulers are also used to enhance training efficiency, ensuring convergence without excessive computational costs.

D. Explainability and Interpretability

To improve the clinical adoption of AI-based glaucoma detection, the system incorporates explainability techniques. Grad-CAM and Class Activation Mapping (CAM) generate heatmaps highlighting the critical regions in retinal images that contribute to the model's decision-making. These visual

explanations overcome the gap between automated detection and human competence by assisting ophthalmologists in understanding AI predictions. Furthermore, the integration of a GPT-based image analysis model allows for textual insights, providing additional context and explanations for each classification result.

E. Deployment and Real-World Integration

The final stage of the system involves deployment in clinical environments. The model is integrated into a web-based or mobile application, enabling ophthalmologists to upload retinal fundus images for automated analysis. A user-friendly interface displays diagnostic results, including prediction confidence scores, heatmap visualizations, and textual explanations. Additionally, the system is designed to support cloud-based computing for scalable and real-time processing, allowing hospitals and clinics to leverage AI-driven diagnostics without requiring high-end local hardware. Future developments aim to integrate multimodal imaging techniques and expand the dataset for improved generalizability and real-world applicability.

IV. METHODOLOGY

The proposed glaucoma detection system follows a structured methodology that integrates deep learning models with explainability and textual analysis techniques to improve diagnostic accuracy and interpretability. The explainability techniques applied to enhance the transparency of the predictions and the GPT-based analysis module for providing textual interpretations of retinal fundus images.

A. Dataset Description

The study utilizes publicly available retinal fundus image datasets, including REFUGE and ORIGA, which contain labeled images of glaucomatous and non-glaucomatous eyes. The dataset is balanced to ensure model generalization and mitigate bias. All input images are standardized to fixed spatial dimensions of 224×224 pixels during preprocessing, ensuring compatibility with the input requirements of diverse network architectures [13].

B. Preprocessing Techniques

Several preprocessing techniques are used to enhance image quality and enable efficient feature extraction.

- **Grayscale Conversion:** The images are converted from RGB to grayscale to emphasize texture-based features relevant to glaucoma detection.
- **Contrast Enhancement:** AHE is used to enhance the visibility of retinal structures.
- **Data Augmentation:** Random rotations, horizontal flips, zooming, and brightness adjustments are used to improve model robustness.

C. Deep Learning Model Selection

For robust feature extraction from retinal fundus images, we employ three popular CNN architectures—ResNet50, VGG16, and EfficientNet. These models, pretrained on large datasets like ImageNet, benefit from transfer learning, enabling faster convergence and improved accuracy with limited labeled data.

1) *ResNet50: Deep Residual Learning:* ResNet50 is a deep convolutional neural network architecture with 50 layers, specifically engineered to mitigate gradient attenuation in models through identity shortcut pathways. These bypass connections allow unimpeded information flow between layers by transmitting original input features forward, enabling stable gradient propagation during backpropagation and reducing training instability in deep architectures. The residual function is formulated as:

$$y = F(x) + x \quad (1)$$

where x is the input feature and $F(x)$ is the residual mapping to be learned. By enabling identity mappings, ResNet50 facilitates the training of deep networks while retaining strong feature representations.

2) *VGG16: Uniform and Deep Feature Extraction:* VGG16 is a 16-layer CNN known for its homogeneous architecture, where all convolutional layers use 3×3 filters. This consistency in filter size ensures deep hierarchical feature extraction while maintaining computational efficiency. Despite its depth, VGG16 is relatively simple compared to more modern architectures but remains effective due to its well-structured convolutional blocks. The sequential nature of VGG16 makes it an interpretable choice for feature extraction in medical imaging tasks.

3) *EfficientNet: Optimized Scaling for Computational Efficiency:* EfficientNet is an advanced CNN architecture designed with a compound scaling approach, which simultaneously scales channel width, layer depth, and input resolution using a set of balanced coefficients. Compared to traditional architectures, EfficientNet provides superior accuracy with fewer parameters, making it highly suitable for medical image analysis where efficiency is a priority.

By integrating these three models, we aim to leverage the strengths of each—ResNet50's deep residual learning, VGG16's structured convolutional approach, and EfficientNet's optimized scalability—to extract rich and meaningful features from retinal fundus images, ultimately enhancing classification performance.

D. Hyperparameter Optimization

To maximize the performance of the selected deep learning models, we employ systematic hyperparameter tuning. Optimizing these parameters ensures improved convergence, prevents overfitting, and enhances generalization to unseen data. The key hyperparameters considered include:

1) *Learning Rate:* The learning rate determines how large the steps are in gradient descent optimization, playing a critical role in whether and how quickly the model converges. We experiment with the following values:

$$\eta \in \{0.001, 0.0001, 0.00001\}$$

2) *Batch Size:* The amount of training samples processed prior to updating the model weights is determined by the batch size. We investigate:

$$B \in \{16, 32, 64\}$$

- **Smaller batch sizes (e.g., 16):** Provide more frequent updates but may introduce variance, leading to noisy optimization.

- Larger batch sizes (e.g., 64): Offer smoother updates but may require more memory and risk poor generalization.

3) *Grid Search and Cross-Validation*: To systematically explore hyperparameter combinations, we utilize grid search combined with k-fold cross-validation. This ensures a comprehensive evaluation across multiple configurations, mitigating the risk of model selection bias. The best hyperparameter set is chosen based on validation accuracy and loss trends. By fine-tuning these hyperparameters, we enhance the robustness and efficiency of our deep learning models, ensuring optimal performance on retinal fundus image classification tasks.

E. Explainability Techniques

1) *Grad-CAM: Gradient-weighted Class Activation Mapping*: Grad-CAM is an effective method for generating heatmaps that highlight the regions of an input image that most impact the model's decision. Unlike conventional CAM, Grad-CAM does not require any architectural changes, making it compatible with various convolutional neural networks (CNNs). The core concept of Grad-CAM involves computing the gradient of the predicted class score z^c with respect to the feature maps B^m in a convolutional layer. The significance of each feature map is determined by the global average of these gradients, which is expressed as:

$$\beta_m^c = \frac{1}{N} \sum_u \sum_v \frac{\partial z^c}{\partial B_{uv}^m} \quad (2)$$

where:

B_{uv}^m represents the activation at spatial position (u,v) in feature map m.

β_m^c is the importance weight for feature map m.

N is the total number of pixels in the feature map.

Once these importance weights are computed, the final class-specific localization map L^c is obtained as:

$$L^c = \sum_k \alpha_k^c A^k \quad (3)$$

Where w^c represents the learned weights associated with class c for feature map A^k . Unlike Grad-CAM, which uses gradients to determine importance weights post-training, CAM directly utilizes the network's learned parameters, making it inherently more interpretable but limiting its applicability to models with a GAP layer. These techniques are particularly useful in medical diagnostics, where model transparency is essential for clinical decision-making. By integrating these explainability techniques, we improve model interpretability, foster trust, and enable better human-AI collaboration in medical imaging applications.

F. GPT-Based Image Analysis

A GPT-based model is integrated to automatically generate meaningful textual explanations from extracted image features, improving interpretability and user accessibility. By leveraging transformer architectures, the model produces context-aware diagnostic descriptions that aid both medical

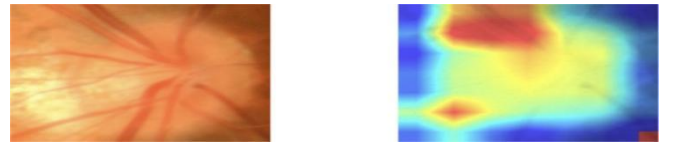
professionals and non-experts. The pretrained GPT model is fine-tuned on domain-specific image-text datasets annotated by medical experts to ensure clinical relevance. Structured prompting and expert feedback further refine the model, enabling accurate identification of abnormalities and clear interpretation of pathological patterns. The GPT-based model does not directly process raw images but instead relies on feature vectors extracted from convolutional neural networks (CNNs). The integration follows a multi-step process: CNNs first extract features from medical images, which are then converted into structured representations by a transformer-based encoder. These structured representations serve as input to the GPT model, which generates a natural language report highlighting key observations. This pipeline ensures that the generated insights are grounded in relevant, high-level image features while maintaining interpretability and coherence.

CAM requires a global average pooling (GAP) layer before the classification layer, making it applicable only to specific CNN architectures. The method works by computing the weighted sum of activation maps directly before classification, defined as:

$$L^c = \sum_k w_k^c A^k \quad (4)$$

where w_k^c represents the learned weights associated with class c for feature map A^k . Unlike Grad-CAM, which uses gradients to determine importance weights post-training, CAM directly utilizes the network's learned parameters, making it inherently more interpretable but limiting its applicability to models with a GAP layer.

Fig. 2. Grad-CAM Image



G. Evaluation Metrics

Critical nature of healthcare applications, these metrics provide a comprehensive assessment of model performance, helping to balance sensitivity and specificity in disease detection. Accuracy is used to gauge overall model correctness, while precision and recall are particularly important in reducing false positives and false negatives, which directly impact clinical decision-making. The F1-score ensures a balanced evaluation, especially when dealing with imbalanced datasets common in medical imaging. Furthermore, ROC-AUC is used to assess the model's capacity to distinguish between pathological and normal cases. By leveraging these evaluation metrics, we can systematically refine our models, improving diagnostic accuracy and ensuring that AI-assisted analysis aligns with real-world clinical requirements.

H. Implementation Details

The models are implemented using TensorFlow and PyTorch. Training is conducted on an NVIDIA GPU, with early stopping applied to prevent overfitting. The final model is deployed as a web-based diagnostic tool, allowing clinicians to upload retinal images and receive AI-generated diagnoses along with visual explanations. Figure 3 represents the comparison chart of various deep learning models and their accuracies. Figure 4 represents loss and accuracy curve on 200 epochs.

Table 1. Comparison between various Deep Learning models

Algorithm	With AdamW	Without AdamW
ResNet50	85%	82%
VGG16	83%	80%
EfficientNet	88%	85%
Custom CNN (Proposed)	92%	89%

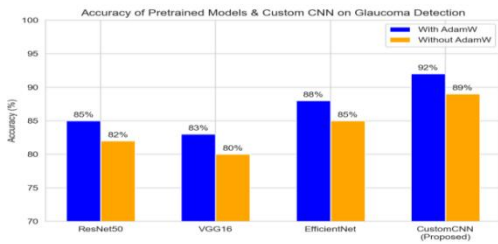


Fig. 3. Comparison Chart between various Deep Learning models

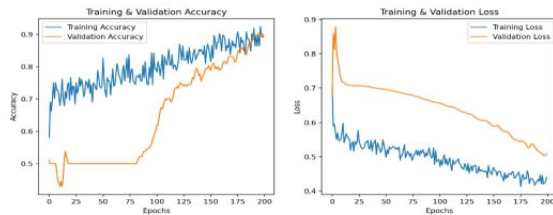


Fig. 4. Accuracy and Loss Curve

V. CONCLUSION

An AI-driven medical image analysis system that integrates deep learning and generative models to enhance diagnostic accuracy and accessibility. By leveraging CNN architectures for feature extraction and a GPT-based model for automated textual explanations, the system provides interpretable insights for both medical professionals and non-expert users. Explainability techniques such as Grad-CAM and CAM improve transparency, while standard evaluation metrics ensure reliability. This method shows how AI can be used in medical diagnostics, opening the door to more precise, understandable, and approachable healthcare solutions.

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